Group Homework #1
Making Good Loans

In this HW, you will use a lending dataset to find the best model to predict who should be given a loan, i.e., determine who will repay in full (or some other criteria you may choose for a good loan). This classification problem assigns a "1" to good loans and "0" to bad loans. The idea is to build a model that will decide on whether to make the loan or not, so you cannot use information that will arrive after the loan is made.

The data file is called "loan.csv."

You will need to decide on the definition of "1" and "0" based on the "loan_status" variable in the dataset. You will want to **exclude** "Current" loans from this analysis as they are not completed loans. Look carefully at the categories and decide which ones you wish to use as 1s and which ones are 0s. This may result in your eliminating rows in your final dataset if you decide not to use all the categories (the ones with very few loans may not be worth retaining).

Carefully design your feature set. You do not need to use all the variables (some may not be useful, such as "id", "member_id", "emp_title", why?). Some of the variables are categorical and will need to be handled carefully. You can normalize and rebalance your data. Intermediate unsupervised learning techniques may or may not help.

There is missing data, you can decide on how you will handle those cases (e.g. impute with an average, or drop rows). Try to drop columns with very few values.

Keep 30% of the data for testing, 70% for training.

You must only use features (variables/columns) in the dataset that are available at the time of making the loan. This will simplify your dataset, of course.

Then use any, some, or all of the classifiers you have learned to construct a good ML model to get as high prediction performance as you can, in terms of AUC (https://scikit-learn.org/stable/modules/model_evaluation.html#scoring-parameter: use the option scoring = 'roc_auc'). You may even ensemble the models.

Your grade will be based on relative AUC versus your peers, so this will be like a competition. Since you have now learned a range of classifiers and other techniques, every team will have a very different solution, which will be interesting. But it will also give you a feel of a real-world ML situation in which you work to find better models.

Extra Background Information

Background on Peer-to-Peer Lending

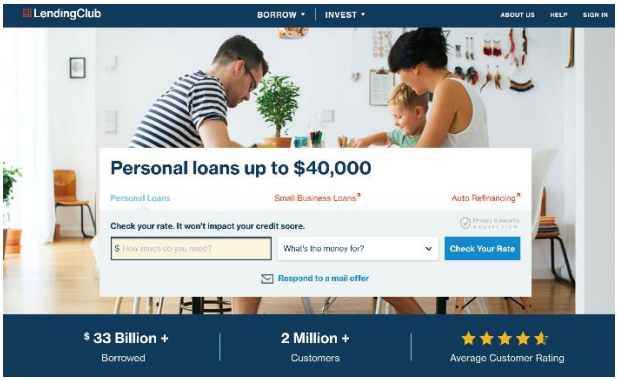
Peer-to-peer lending refers to the practice of lending money to individuals (or small businesses) via online services that match anonymous lenders with borrowers. Lenders can typically earn higher returns relative to savings and investment products offered by banking institutions. However, there is of course the risk that the borrower defaults on his or her loan.

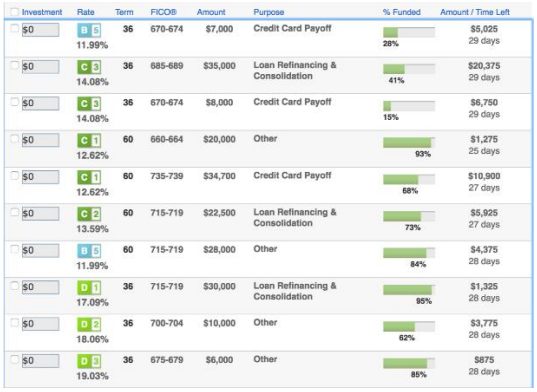
Interest rates are usually set by an intermediary platform on the basis of analyzing the borrower's credit (using features such as FICO score, employment status, annual income, debt-to-income ratio, number of open credit lines). The intermediary platform generates revenue by collecting a one-time fee on funded loans (from borrowers) and by charging a loan servicing fee to investors.

The peer-to-peer lending industry in the U.S. started in February 2006 with the launch of Prosper, followed by LendingClub. In 2008, the Securities and Exchange Commission (SEC) required that peer-to-peer companies register their offerings as securities, pursuant to the Securities Act of 1933. Both Prosper and LendingClub gained approval from the SEC to offer investors notes backed by payments received on the loans.

By June 2012, LendingClub was the largest peer-to-peer lender in the U.S. based on issued loan volume and revenue, followed by Prosper. In December 2015, LendingClub reported that $15.98 billion in loans had been originated through its platform. With very high year-over-year growth, peer-to-peer lending has been one of the fastest growing investments. According to InvestmentZen, as of May 2017, the interest rates range from 6.7%-22.8%, depending on the loan term and the rating of the borrower, and default rates vary between 1.3% and 10.6%.(http://www.investmentzen.com/peer-to-peer-lending-for-investors/lendingclub} (date accessed)

LendingClub issues loans between $1,000 to $40,000 for a duration of either 36 months or 60 months. As mentioned, the interest rates for borrowers are determined based on personal information such as credit score and annual income. In addition, LendingClub categorizes its loans using a grading scheme (grades A, B, C, D, E, and F, where grade A corresponds to the loans judged to be ``safest'' by LendingClub). Individual investors can browse loan listings online before deciding which loans(s) to invest in:

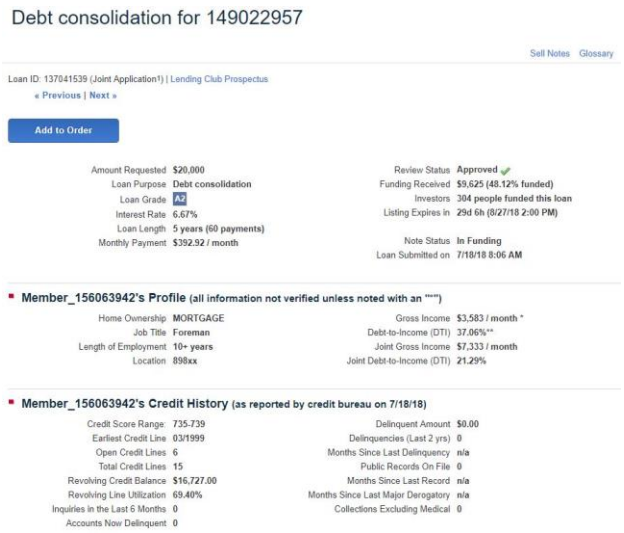




Investment	Rate	Term	FICO®	Amount	Purpose	% Funded	Amount / Time Left
$0	B 5 11.99%	36	670-674	$7,000	Credit Card Payoff	28%	$5,025 29 days
$0	C 3 14.08%	36	685-689	$35,000	Loan Refinancing & Consolidation	41%	$20,375 29 days
$0	C 3 14.08%	36	670-674	$8,000	Credit Card Payoff	15%	$6,750 29 days
$0	C 1 12.62%	60	660-664	$20,000	Other	93%	$1,275 25 days
$0	C 1 12.62%	60	735-739	$34,700	Credit Card Payoff	68%	$10,900 27 days
$0	C 2 13.59%	60	715-719	$22,500	Loan Refinancing & Consolidation	73%	$5,925 27 days
$0	B 5 11.99%	60	715-719	$28,000	Other	84%	$4,375 28 days
$0	D 1 17.09%	36	715-719	$30,000	Loan Refinancing & Consolidation	95%	$1,325 28 days
$0	D 2 18.06%	36	700-704	$10,000	Other	62%	$3,775 28 days
$0	D 3 19.03%	36	675-679	$6,000	Other	85%	$875 28 days

Each loan is split into multiples of $25, called notes (for example for a $2,000 loan, there will be 80 notes of $25 each). Investors can obtain more detailed information on each loan by clicking

on the loan. The Figure below shows an example of the additional information available for a given loan. Investors can then purchase these notes in a similar fashion to ``shares'' of a stock in an equity market. Of course, the safer the loan the lower the interest rate, and so investors have to balance risk and return when deciding which loans to invest in.



Debt consolidation for 149022957

Sell Notes Glossary

Loan ID: 137041539 (Joint Application1) | Lending Club Prospectus

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Add to Order

Amount Requested	$20,000	Review Status	Approved
Loan Purpose	Debt consolidation	Funding Received	$9,625 (48.12% funded)
Loan Grade	A2	Investors	304 people funded this loan
Interest Rate	6.67%	Listing Expires in	29d 6h (8/27/18 2:00 PM)
Loan Length	5 years (60 payments)		
Monthly Payment	$392.92 / month	Note Status	In Funding
		Loan Submitted on	7/18/18 8:06 AM

Member_156063942's Profile (all information not verified unless noted with an "*")

Home Ownership	MORTGAGE	Gross Income	$3,583 / month *
Job Title	Foreman	Debt-to-Income (DTI)	37.06%**
Length of Employment	10+ years	Joint Gross Income	$7,333 / month
Location	898xx	Joint Debt-to-Income (DTI)	21.29%

Member_156063942's Credit History (as reported by credit bureau on 7/18/18)

Credit Score Range	735-739	Delinquent Amount	$0.00
Earliest Credit Line	03/1999	Delinquencies (Last 2 yrs)	0
Open Credit Lines	6	Months Since Last Delinquency	n/a
Total Credit Lines	15	Public Records On File	0
Revolving Credit Balance	$16,727.00	Months Since Last Record	n/a
Revolving Line Utilization	69.40%	Months Since Last Major Derogatory	n/a
Inquiries in the Last 6 Months	0	Collections Excluding Medical	0
Accounts Now Delinquent	0		

Data Description

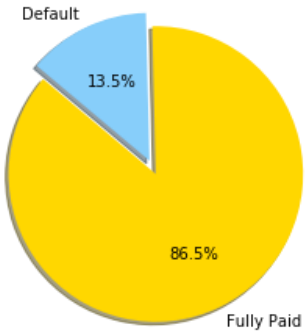
The dataset contain comprehensive information on all loans issued between 2007 and the third quarter of 2017 (a new updated dataset is uploaded every quarter). The data records hundreds of features including the following, for each loan:

- Interest rate,
- Loan amount,
- Monthly installment amount,
- Loan status (e.g., fully-paid, default, charged-off),
- Several additional attributes related to the borrower such as type of house ownership, annual income, monthly FICO score, debt-to-income ratio, and number of open credit lines.

The definition of each loan status is summarized in the below table:

Number of Days Past Due	Status
0	Current
16-120	Late
121-150	Default
150+	Charged-Off

- Current refers to a loan that is still being reimbursed in a timely manner.
- Late corresponds to a loan on which a payment is between 16 and 120 days overdue.
- If the payment is delayed by more than 121 days, the loan is considered as being in Default.
- If LendingClub has decided that the loan will not be paid off, then it is given the status of Charged-Off. Note that sometimes the ``Charged-Off'' status will occur before ``Default'' if/when the borrower has filed bankruptcy or has notified the intermediary platform.

These dynamics imply that five months after the term of each loan has ended, every loan ends in one of two LendingClub states -- fully paid or charged-off. For example, if a borrower defaults on a loan in the last month of a 36-month loan, it would take another five months for the loan to be charged-off.

A significant portion (13.5%) of loans ended in Default status; depending on how much of the loan was paid back, these loans might have resulted in a significant loss to investors who had invested in them. The remainder were Fully Paid -- the borrower fully reimbursed the loan's outstanding balance with interest, and the investor earned a positive return on his or her investment. Therefore, to avoid unsuccessful investments, our goal is to estimate which loans are more (resp. less) likely to default (or get charged-off) and which will yield low (resp. high) returns.